PORTFOLIO ENGINE // ARCHITECTURE & CODE MANUAL

COMPLETE SYSTEMS BLUEPRINT • DEVELOPER CONTROL GUIDE

STATUS: v5.4.4-SHIPPING
STACK: HTML5 / CSS3 / JS / CF WORKER

A comprehensive, color-coded reference manual for Mayank Manjul's portfolio codebase (mayankmanjul.com). Covers project layout, editing guidelines, local previewing, and Cloudflare automated deployment.

01 // RECOMMENDED IDE ENVIRONMENT

JetBrains Rider

RECOMMENDED FOR C++/UNITY

Because you are an Unreal Engine & Unity developer, **JetBrains Rider** is your optimal environment. Rider embeds the entire **WebStorm engine** natively:

- **Zero configuration**: Full CSS variable autocomplete, HTML5 tag inspection, and JavaScript navigation.
- **Git Integration**: World-class visual changelist diffing and branch management.
- **How to open**: Open Rider → *File* → *Open* → select folder `D:\Project\portfolio-engine`.

VS Code + Live Server

BEST FOR HOT-RELOAD

Ultra-lightweight industry standard for modern web programming with 1-second startup time:

- **Install Extension**: *"Live Server"* (by Ritwick Dey).
- **Instant Preview**: Right-click `index.html` → click **"Open with Live Server"**.
- **Hot-Reload**: Any edit saved in HTML, CSS, or JS immediately refreshes in your browser.

02 // REQUEST LIFECYCLE & DATA FLOW

```
[1. VISITOR CLIENT] Browser requests https://mayankmanjul.com/
        ▼
[2. CLOUDFLARE EDGE] Worker Router (worker.js) serves cached static assets (index.html, styles/*, scripts/*) via env.ASSETS
        ▼
[3. CLIENT RUNTIME] Browser executes HUD Telemetry, Web Audio Synthesizer, and Netcode Simulation Canvas
        ▼
[4. CONTACT TRANSMISSION] User clicks "Dispatch Secure Transmission" → JavaScript issues fetch('/api/contact')
        ▼
[5. RESEND API DISPATCH] worker.js validates payload, authenticates via RESEND_API_KEY, dispatches email to mayankmanjul@live.com
```

Zero Desktop Mail Client Popups: Inquiries are transmitted silently over HTTPS edge APIs. The visitor never leaves your site, and Outlook / desktop mail apps are completely bypassed.

03 // FILE-BY-FILE CODEBASE BLUEPRINT

FILE / PATH	DOMAIN	PURPOSE & WHERE TO MAKE CHANGES
`index.html`	MARKUP	**Master Page Backbone**. Contains all 8 HUD sections, project cards, metric badges, SVG icons, resume links, and modal blueprint templates.
`styles/main.css`	DESIGN SYSTEM	**Global CSS Tokens** (`:root` colors, typography scales, container max-widths, high-contrast `:focus-visible` rings, and accessibility skip links).
`styles/hud-theme.css`	THEMES & FX	**Cyberpunk Aesthetic**. Controls 4 active color themes (Slate Cyan, Amber Gold, Matrix Green, Crimson Red), scanlines, glows, and `prefers-reduced-motion` .
`styles/components.css`	COMPONENTS	**Modular UI Panels**. Radar chart frame, shipped project grid cards, experience timeline items, contact cards, and technical spec dialog modals.
`styles/mobile.css`	MOBILE ⩽768px	**Mobile Optimized Viewport**. Converts multi-column desktop grids into smooth, battery-efficient accordion decks and disables heavy canvas particles.

04 // JAVASCRIPT LOGIC CONTROLLERS (SCRIPTS/)

scripts/main.js — CORE CONTROLLER

The central orchestrator of interactive HUD events:

- **Uptime & Telemetry**: Calculates live uptime clock and simulated frame time.
- **Scrollspy**: Detects section scroll position and updates top nav indices (`01` to `08`).
- **Theme Engine**: Toggles body theme class and saves user selection to `localStorage`.
- **Contact API Dispatch**: Collects form data and issues asynchronous `fetch('/api/contact')`.

scripts/sound-fx.js — AUDIO SYNTHESIZER

Procedural Web Audio API sound generator:

- **Zero Audio Files**: Synthesizes custom sine and square waves at runtime (tactile clicks, laser chimes).
- **Safe Browsing**: Respects browser autoplay restrictions; remains muted until the user clicks the top bar audio toggle.

scripts/physics-sandbox.js — NETCODE SIMULATION

Interactive 2D physics & netcode lab:

- Demonstrates client-side prediction, server authoritative reconciliation, jitter buffers, and simulated packet loss.

worker.js & build.js — CLOUDFLARE EDGE

Serverless Edge Router & deployment hook:

- **`worker.js`**: Routes `/api/contact` and falls back to static files.
- **`build.js`**: Automatically injects `RESEND_API_KEY` during Cloudflare deploy while keeping GitHub repository clean.

05 // LOCAL DEVELOPMENT & GIT DEPLOYMENT WORKFLOW

Running Locally (5 Seconds) — DEV WORKFLOW

No npm installs or build steps needed. Run directly in PowerShell:

```
python -m http.server 8080
```

Then open `http://localhost:8080` in Chrome to test changes in real time.

Pushing Updates to Production — DEPLOYMENT

Deploying changes to `mayankmanjul.com` is a standard 3-command git workflow:

```
git add .
git commit -m "feat: update project
details"
git push origin main
```

Cloudflare automatically detects the push, runs `build.js`, and publishes the update live in ~60 seconds.

06 // QUICK EDIT CHEAT SHEET

WHAT YOU WANT TO UPDATE	FILE LOCATION	WHAT TO EDIT
Update Resume PDF URL	`index.html`	Search for `drive.google.com` and update the Google Drive URL on lines 152, 275, and 1862.
Add / Edit Shipped Game	`index.html`	Search for `id="projects"` (Line ~345). Copy any `<article class="project-card">` block to add a new game.
Edit Career Experience	`index.html`	Search for `id="experience"` (Line ~1300). Edit role titles, company names, or bullet points.
Customize Theme Colors	`styles/hud-theme.css`	Look at lines 1–60. Adjust the hex values for `--theme-accent` and `--theme-accent-secondary`.
Change Form Target Email	`worker.js`	Search for `to: ["mayankmanjul+personal@live.com"]` on line ~51.

🔒 **Security Rule:** Never hardcode API keys or secrets into git files. Always manage private tokens inside the Cloudflare Dashboard under *Settings → Variables and secrets*.